
Structural and dynamical properties of liquid water using neural network potentials

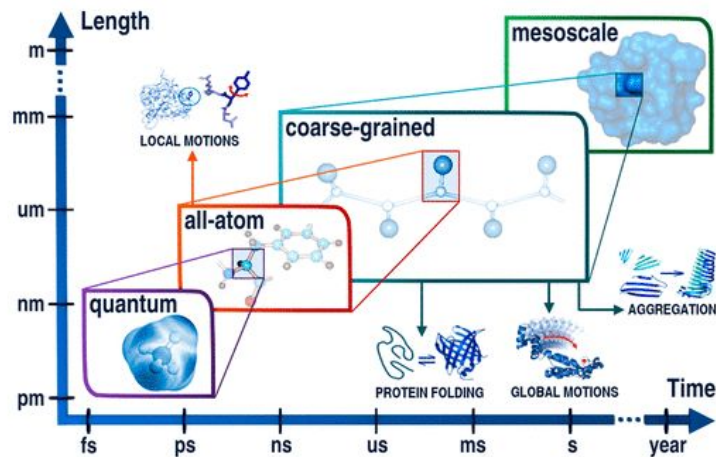
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CHEM 31300
Machine Learning in Chemistry

Introduction

Neural Network Potentials (NNPs): A neural network based representation of the potential energy surface (PES) of the system

- Ab initio molecular dynamics (AIMD):
 - Explicit quantum chemical calculations
 - Accurate (but very slow)
- Classical molecular dynamics (MD):
 - Empirical potentials (force fields)
 - Fast (but lacks electronic information)
- Neural network potentials (NNPs):
 - To achieve a balance between speed and accuracy



NNPs

Machine-learned interatomic potentials bridging the gap between speed and quantum chemical accuracy

Generate training data (DFT, etc.)



Train NNPs:

- Feature representation: descriptors for symmetry invariance/equivariance
- Atom-centered symmetry functions, SOAP, graph-based descriptors, etc
- Loss function typically uses energies, forces and stress tensors



Deploy for MD simulations

Some popular architectures:

- **Behler-Parrinello NNs:**
 - Atom-centered descriptors

$$E = \sum_i E_i = \sum_i E(\mathcal{R}^i)$$

- **Deep potential MD (DeePMD):**
 - Atom-centered
- **MACE:**
 - Equivariant message passing
- Others: NequIP, Allegro, SchNet, etc.

A perfect water model?

Water: Critically important solvent!

- Really difficult to construct a *perfect* water model!
 - Complex hydrogen bonding
 - Electrostatic and polarization effects
 - Nuclear quantum effects
 - Complex phase behavior at supercooled temperatures
- Existing models: captures a few things but not everything(?)
 - Empirical models: SPC, TIP3P, etc
 - Data-driven models: MB-pol, NNP-based water models
 - MB-pol: many-body polarizable model (*really good!*)

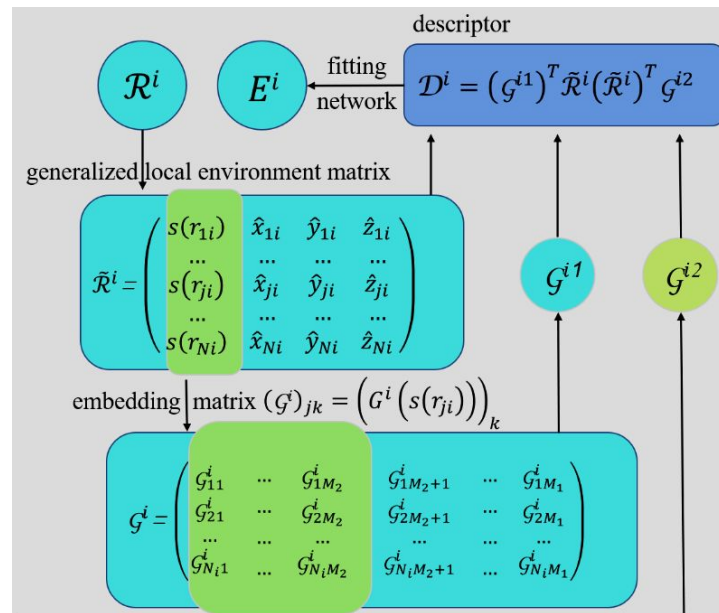
PROJECT GOAL: Critically evaluating the ability of NNP-based water models to capture the structure and dynamics of water

DNN@MB-pol (DeePMD variant)

- DeePMD framework:

$$E = \sum_i E_i = \sum_i E(\mathcal{R}^i)$$

- DNN@MB-pol
 - DeePMD potential trained on MB-pol generated water trajectories
 - Trained on ~3000 configurations across 300–500 K and 1–1000 bar
 - Active learning used
- Merits and demerits:
 - Accurate prediction of liquid water properties
 - Incorrect prediction of liquid-vapor equilibrium



MACE

$$\sigma_i^{(t)} = (\mathbf{r}_i, z_i, \mathbf{h}_i^{(t)}) \longrightarrow m_i^{(t)} = \sum_j u_1(\sigma_i^{(t)}; \sigma_j^{(t)}) + \sum_{j_1, j_2} u_2(\sigma_i^{(t)}; \sigma_{j_1}^{(t)}, \sigma_{j_2}^{(t)}) + \cdots + \sum_{j_1, \dots, j_\nu} u_\nu(\sigma_i^{(t)}; \sigma_{j_1}^{(t)}, \dots, \sigma_{j_\nu}^{(t)})$$

$$h_i^{(t+1)} = U_t(\sigma_i^{(t)}, m_i^{(t)})$$

$$E_i = \sum_{t=1}^T \mathcal{R}_t(\sigma_i^{(t)})$$

- MACE: Higher order equivariant message passing neural networks
- ACE: Atomic Cluster Expansion - higher order complete polynomial basis functions
- Equivariant GNN
- MACE-MP-0
 - Pre-trained *foundational* NNPs trained on materials datasets

Implementation

Models:

- SPC/Fw: classical force field
- DNN@MB-pol: DeePMD variant of MB-pol
- MACE-MP-0b: trained on MPTrj dataset (PBE+U DFT level of theory)
- MACE-MPA-0: trained on MPTrj+sAlex dataset (PBE+U DFT level of theory)
- MACE-MATPES-r2SCAN-0: trained on MATPES-r2SCAN dataset (r^2 SCAN DFT level of theory)

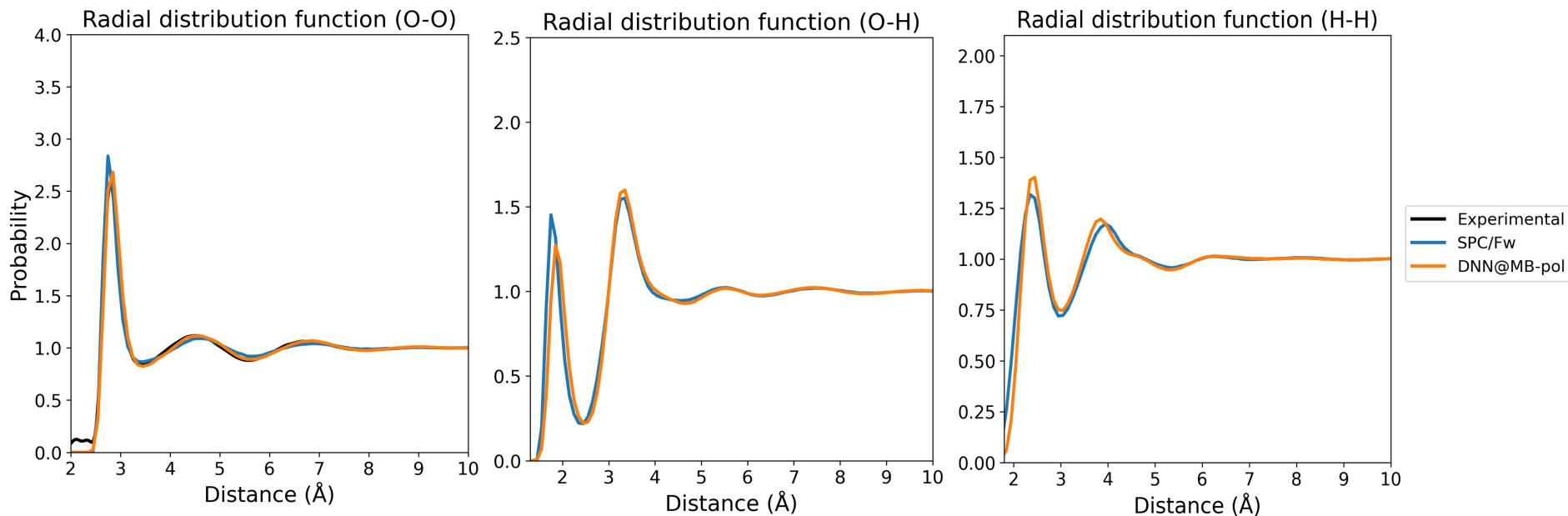
Molecular Dynamics Simulations:

- LAMMPS: patched with DeePMD (Tensorflow) and MACE (PyTorch)
- MACE simulations on CUDA
- NVT runs with Nose-Hoover thermostat
- ~800 ps runs
- Timestep: 1 fs for NNPs, 2 fs for SPC/Fw

Radial distribution functions

RDF: Distribution functions to assess the two-body or pairwise structural correlations of a system

$$g(R) = \frac{V}{N^2} \left\langle \sum_I \sum_{J \neq I} \delta(R - R_{IJ}) \right\rangle$$

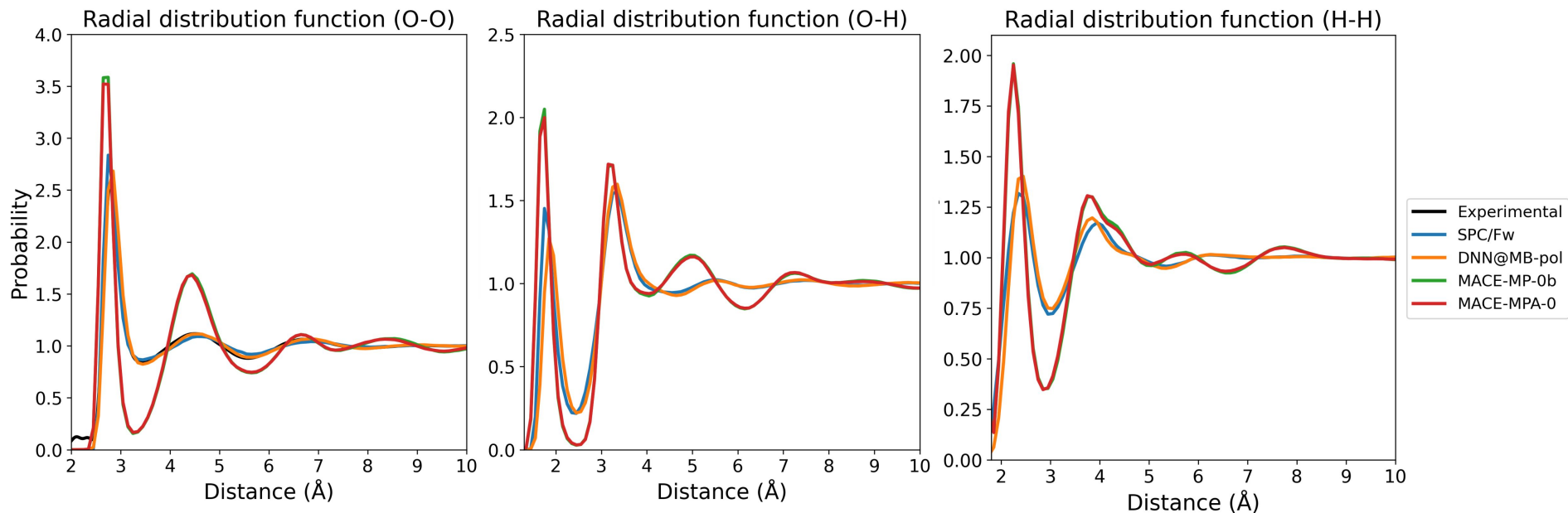


- SPC/Fw and DNN@MB-pol have excellent agreement with each other (and with experimental RDF)

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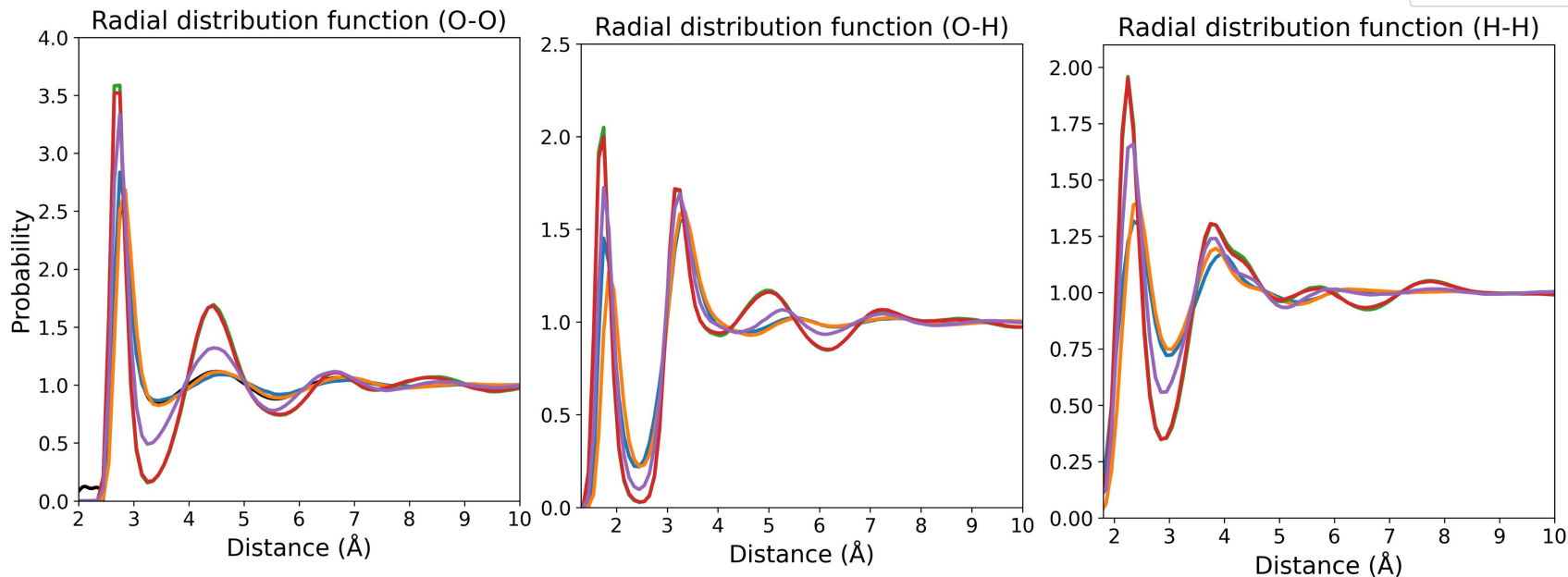


- MACE-MPA-0 and MACE-MP-0b have significant degree of overstructuring

Radial distribution functions

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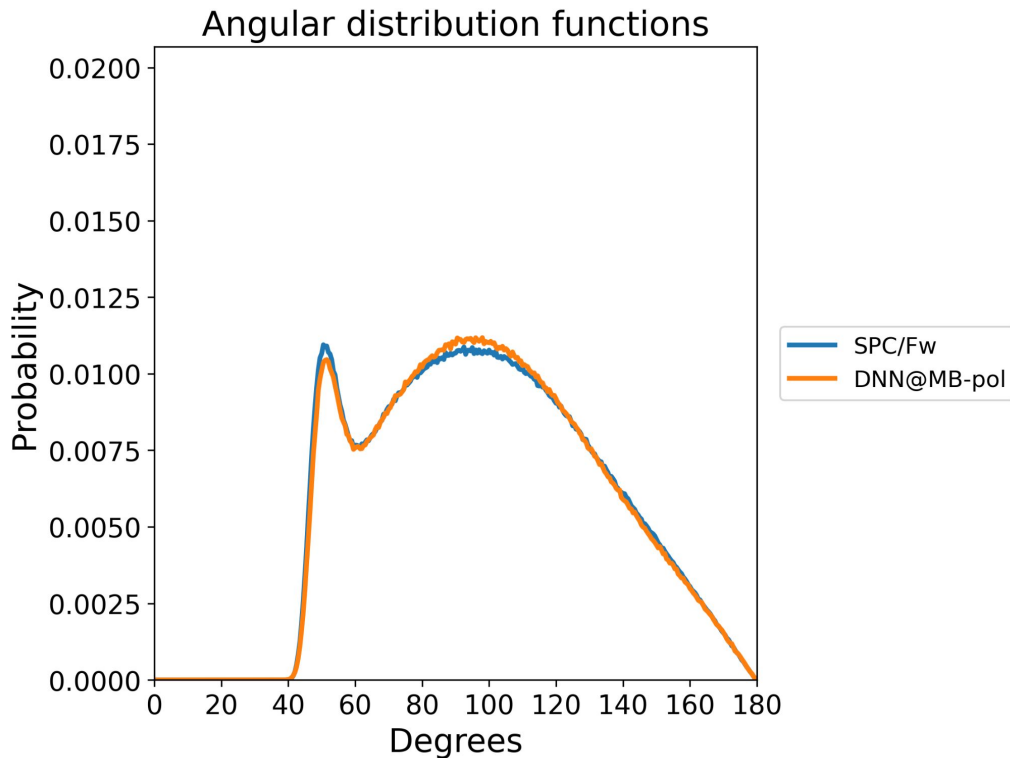
- MACE-MATPES-r2SCAN-0 improves upon MACE-MP models (but not as good as SPC/Fw or DNN@MB-pol)

Angular distribution functions

ADF: Distribution functions to assess the three-body structural correlations of a system

$$P(\theta) = \frac{1}{W} \left\langle \sum_I \sum_{J \neq I} \sum_{K > J} \delta(\theta - \theta_{JIK}) \right\rangle$$

- ADFs computed upto 3.7 Å (first solvation shell of water)
- COM-COM-COM type of triplets considered
- SPC/Fw and DNN@MB-pol have excellent agreement with each other

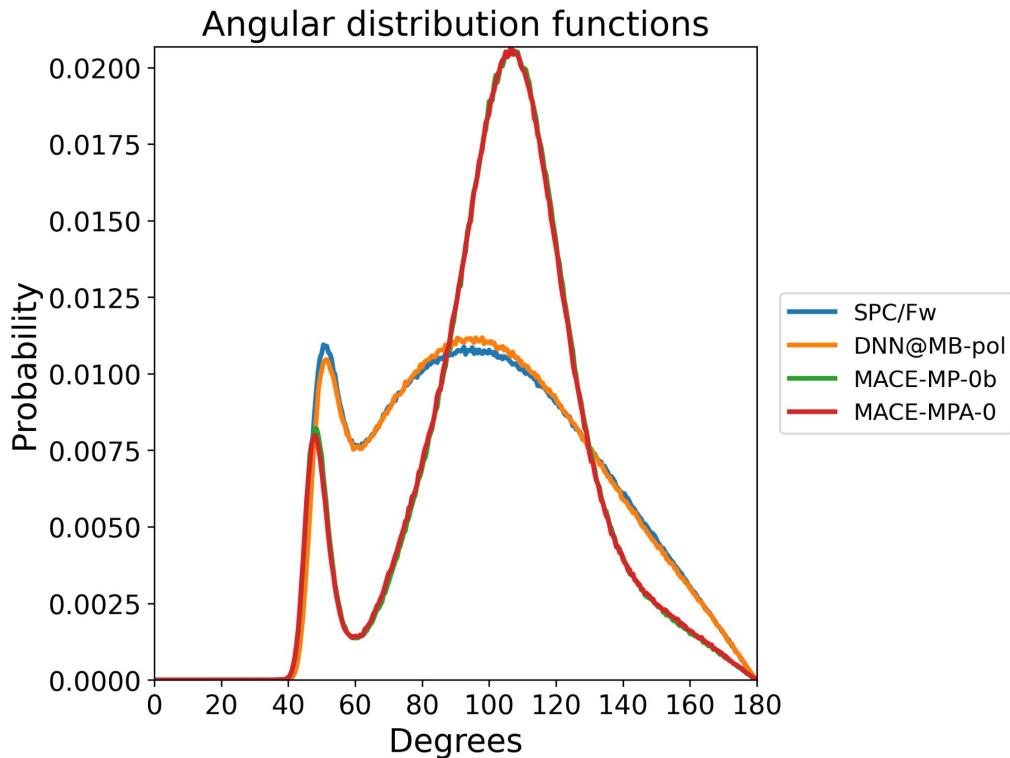


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- COM-COM-COM type of triplets considered
- MACE-MP models cannot capture ADFs

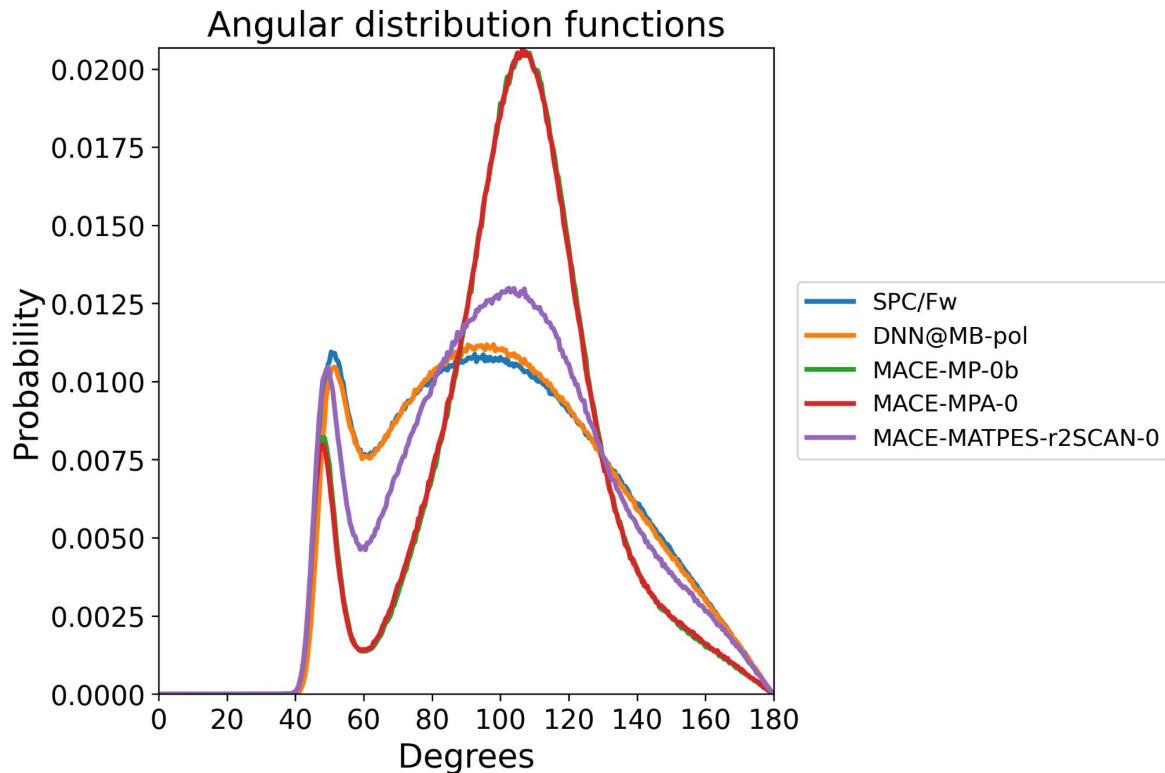


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- ADFs computed upto 3.7 Å (first solvation shell of water)
- COM-COM-COM type of triplets considered
- MACE-MATPES-r2SCAN-0 shows significant improvement

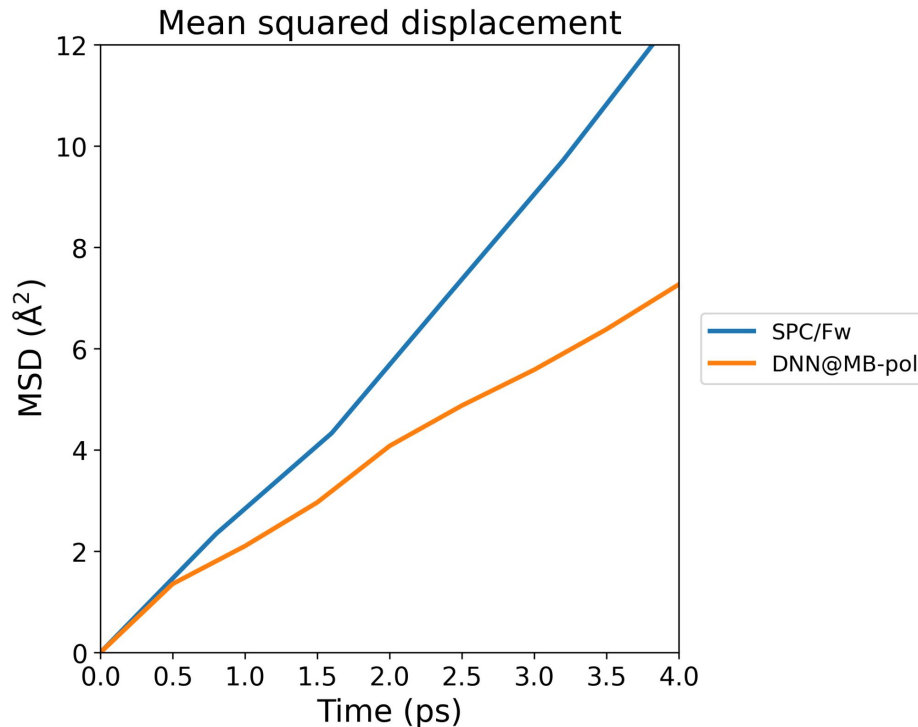


Mean squared displacement

MSD: Metric to assess the dynamical evolution of the atoms of a system (can be used to calculate the self-diffusion coefficient)

$$\text{MSD}(t) = \langle |\mathbf{R}(t) - \mathbf{R}(0)|^2 \rangle$$

- Steeper the curve, more is the diffusion
- DNN@MB-pol predicts lower diffusion than SPC/Fw

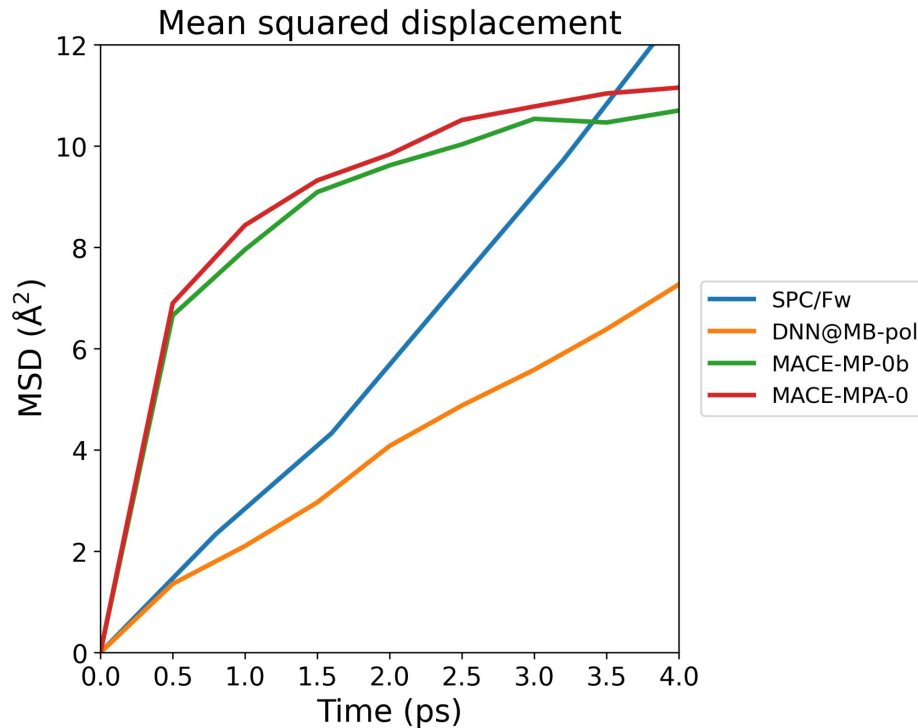


Dynamical properties

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- Steeper the curve, more is the diffusion
- MACE-MP models predict more diffusive behavior

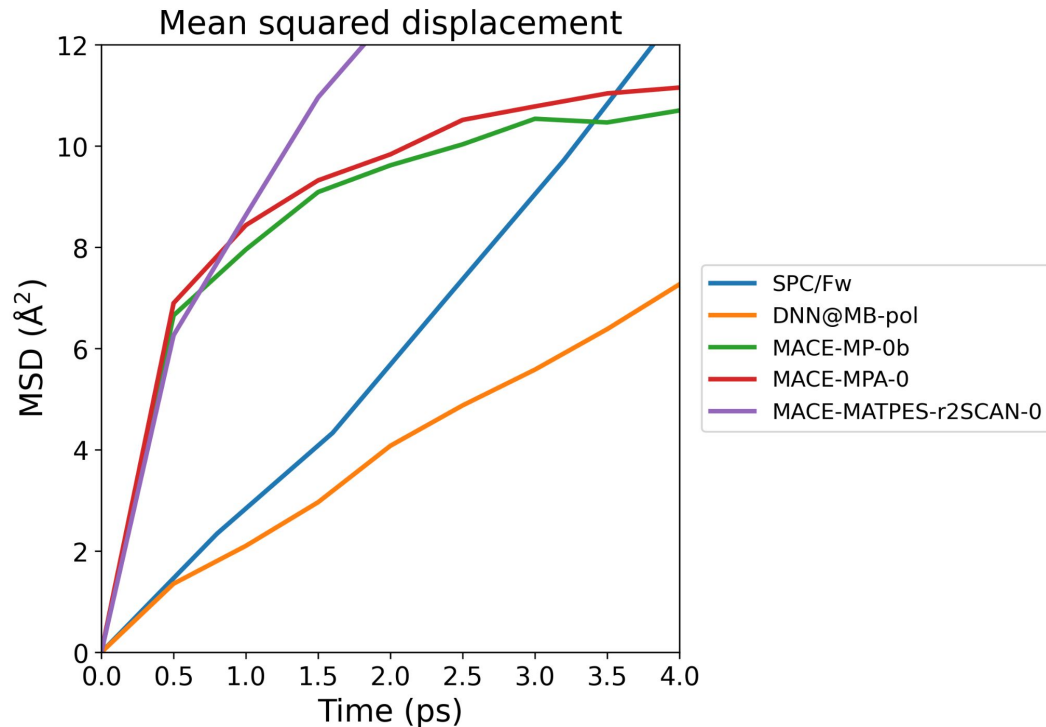


Dynamical properties

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$$\text{MSD}(t) = \langle |\mathbf{R}(t) - \mathbf{R}(0)|^2 \rangle$$

- Steeper the curve, more is the diffusion
- MACE-MATPES-r2SCAN-0 predicts steepest MSD curve, possibly overestimating diffusion



Conclusions and outlook

- DNN@MB-pol predicts highly accurate structural correlations (RDFs and ADFs)
- MACE models predicts overstructuring of water, as well as more diffusive behavior
- However, ML models are as good as the training data
- MACE-MP models trained on PBE (not good for water)

How to improve speed and accuracy?

- Active learning tools can be used for improving foundation models like MACE
- Start with a known model → feed in more data → train again → deploy
- GPU based implementations of equivariant NNPs (typically quite slow)

Questions?